

**An Analysis of Salary's Impact on Job Satisfaction Levels for Software Developers in the
United States**

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Executive Summary

This study analyzes whether yearly salary impacts job satisfaction levels across software developers in the United States. Using data from Stack Overflow Insights, the data was cleaned via Python, grouped into salary-level quartiles, and analyzed to compare satisfaction levels. The results revealed a very weak positive correlation ($r=0.04$), demonstrating that yearly salary has little effect on job satisfaction. However, the study was limited by human error and biases, incomplete responses, and a narrow industry sample. Nonetheless, these findings suggest that non-salary factors-such as work-life balance, positive workplace culture, and a benefits package (retirement, paid time off)-play a larger role in determining job satisfaction levels.

Introduction

Job satisfaction refers to the “pleasurable or positive emotional state, resulting from the appraisal of one’s job experiences (Montuori et al., 2022).” Generally speaking, it is influenced by work-life balance, salary, career benefits, recognition for quality performance, company culture, etc. When an employee has high job satisfaction, they produce higher quality work, develop deeper personal connections with coworkers, and are less at risk of absenteeism or burnout. Thus, companies then see a lower turnover rate and higher financial performance (Shashenkova, 2024).

Many workers develop a preconceived notion that a higher yearly salary leads to higher job satisfaction. However, studies have concluded that yearly salary has little effect on job satisfaction levels. Instead, workplaces with an emphasis on employee well-being, meaningful work, and career advancement have higher employee job satisfaction than companies that offer higher pay (Bolden-Barrett, 2019). This research study intends to take a deeper look into the correlation between yearly pay and job satisfaction in the software development industry. The

purpose of this study is to determine if yearly pay is the only factor that affects employee satisfaction, or if it is a multi-faceted correlation. This paper will explain the data collected and methods used (for data cleaning and analysis), the results and a discussion of the findings (including visual representations), possible limitations, and a conclusion to determine the ultimate implications.

Data and Methods

The data set was sourced from The Stack Overflow Annual Developer Survey 2025, whose goal is to determine the current state of the software development industry across the world by collecting quantitative and qualitative data. The 2025 Survey had “over 49,000 responses from 177 countries across 62 questions focused on 314 different technologies (*Stack Overflow Insights*, n.d.).” The large number of respondents allows for the researchers to accurately establish correlations between variables. For this project, the independent variable is salary level and the dependent variable is job satisfaction.

The researchers chose to use Google Colab and Python for the data wrangling process. The data was filtered to respondents that had the following attributes: United States resident, full-time employee, and 1-30 years of experience. This creates a focused study that establishes a more accurate correlation (by limiting noise). The data was further filtered to salaries between \$30,000 to \$500,000 (removing extreme outliers). Finally, the incomplete data in the “ConvertedCompYearly” was removed to reduce possible biases. This left a total of 4,000 results that maintained integrity and reliability of the study.

Results and Discussion

To determine whether there is a relationship between salary and job satisfaction, it was crucial to apply various data wrangling techniques on the survey data in Python. Initial

descriptive statistics showed the overall distribution of job satisfaction ratings and annual salary levels across the sample as shown in *Figure 1* where the main highlighted variables are displayed. After cleaning the data, a new column was created titled ‘SalaryQuartile’, where Q1 (lowest 25%) had 1,006 respondents, Q2 (25%-50%) had 995 respondents, Q3 (50%-75%) had 1,011 respondents, and Q4 (top 25%) had 990 respondents. These quartiles utilized the ‘JobSat’, ‘ConvertedCompYearly’, and ‘WorkExp’ variables so the information was more specific to the research question. In *Figure 2*, the box plots show evidence, especially in the second and third quartiles, that there is a steady increase in salary median and spread. The first three quartiles are all rather tight boxes, meaning that the salaries are all clustered closely together, while the fourth has far more variability in the higher earning respondents’ salaries and even outliers near 480k and 500k.

The job satisfaction across each quartile was also calculated, and each one has very similar variability. Each salary group has the same range (10) of answers from very satisfied to dissatisfied. According to *Figure 3*, there are some low earners who are satisfied with their job while some higher earners are very dissatisfied and vice versa. The standard deviations or how far the values dispersed from the mean, are almost identical. The stacked histogram shows levels of job satisfaction from low (dark purple) to high (yellow), and high satisfaction levels (7-10) were reported most commonly in every quartile. When looking at the mean of the quartiles, it does show that from Q1 to Q4, there is a 0.37 point increase (7.02 to 7.39). This is a very miniscule amount in the grand scheme of things however.

This evidence all seems to point to the theory that there is a relationship between job satisfaction and salary, but not to the extent that most might think. Before reaching our conclusion, it was necessary to test the correlation by creating a new data frame with the relevant

columns ('ConvertedCompYearly' and 'JobSat'), in order to isolate the data for the correlation calculation to get an accurate result. The correlation coefficient was found to be 0.04 which indicates a very weak but positive relationship between the two variables. Because the value is so close to zero, there is no linear connection and it is safe to refute the question and look to other factors for a relationship between job satisfaction and salary. Based on the findings of the data analysis, it can be inferred that compensation barely affects job satisfaction.

Limitations and Conclusions

After seeing the results, it is also crucial to include constraints for the data as would be in the real world. Limitations represent weaknesses in the data which our group has identified. The most common seems to be the scope of variables. This is a very large data set originally with many duplicate and missing values along with several variables that have nothing to do with the research question. The data needed to be cleaned and wrangled, but even so the survey was filled out by people and that can lead to human error. Selection bias could have taken place since the survey only recorded answers from developers (software, machine learning, etc.) and no other profession. It's not clear how many people were sent the survey but chose not to answer which could lead to response bias.

As stated previously, when an employee has high job satisfaction, they perform better by producing higher quality work and are less at risk of burnout. Our group has concluded that there are other factors that matter more than just salary. The following aspects are what have been deemed the most important to job satisfaction for all professions. A positive work environment allows employees to collaborate and work with one another in a safe space while also creating reliability between coworkers (Walker, 2024). Work-life balance helps create more engagement, better performance, and productivity. This does not mean a split between work and personal life,

but feeling fulfilled through both and ensuring you can set boundaries and manage your time effectively. Rewards and recognition such as pay raises and career growth opportunities are also appreciated in the workplace and shows you recognize the employees hard work and dedication to the company (Walker, 2024).

Recommendations

The investigation into the question, “Is there a relationship between job satisfaction and salary or do other factors matter more,” has revealed the weak correlation between the two. On this basis, our group has determined other factors aside from salary level such as company culture, work-life balance, etc. play a key role in job satisfaction too. Our recommendation to companies that are delving into how to improve job satisfaction, even if they are already making a fair salary, would first be to design a benefits package with perks that appeal to your employees’ priorities. Health and wellness, paid time off, family, and retirement benefits are all grouped into a valuable category and can help employees feel more secure and supported (Walker, 2024). As far as work-life balance goes, employers can experiment with flexible work schedules to positively impact their company whether that be through remote or hybrid days. These schedules allow employees to save time on commuting to work and also allow for more time throughout the day to be productive. When an employer supports these aspects, it helps foster a positive company culture.

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Appendix

Figure 1: Descriptive Statistics Table

	count	mean	std	var	min	25%	mode	50%	75%	max	range	iqr
SalaryQuartile												
Q1 (lowest 25%)	708.0	7.019774	2.177331	4.740768	0.0	6.0	8.0	7.0	8.0	10.0	10.0	2.0
Q2 (25%-50%)	849.0	7.261484	1.999985	3.999942	0.0	6.0	8.0	8.0	9.0	10.0	10.0	3.0
Q3 (50%-75%)	863.0	7.217845	2.051836	4.210029	0.0	6.0	8.0	8.0	9.0	10.0	10.0	3.0
Q4 (top 25%)	836.0	7.391148	2.005709	4.022868	0.0	7.0	8.0	8.0	9.0	10.0	10.0	2.0

Figure 2: Salary Distribution by Quartile in USA



Figure 3: Job Satisfaction Levels by Salary Quartile

